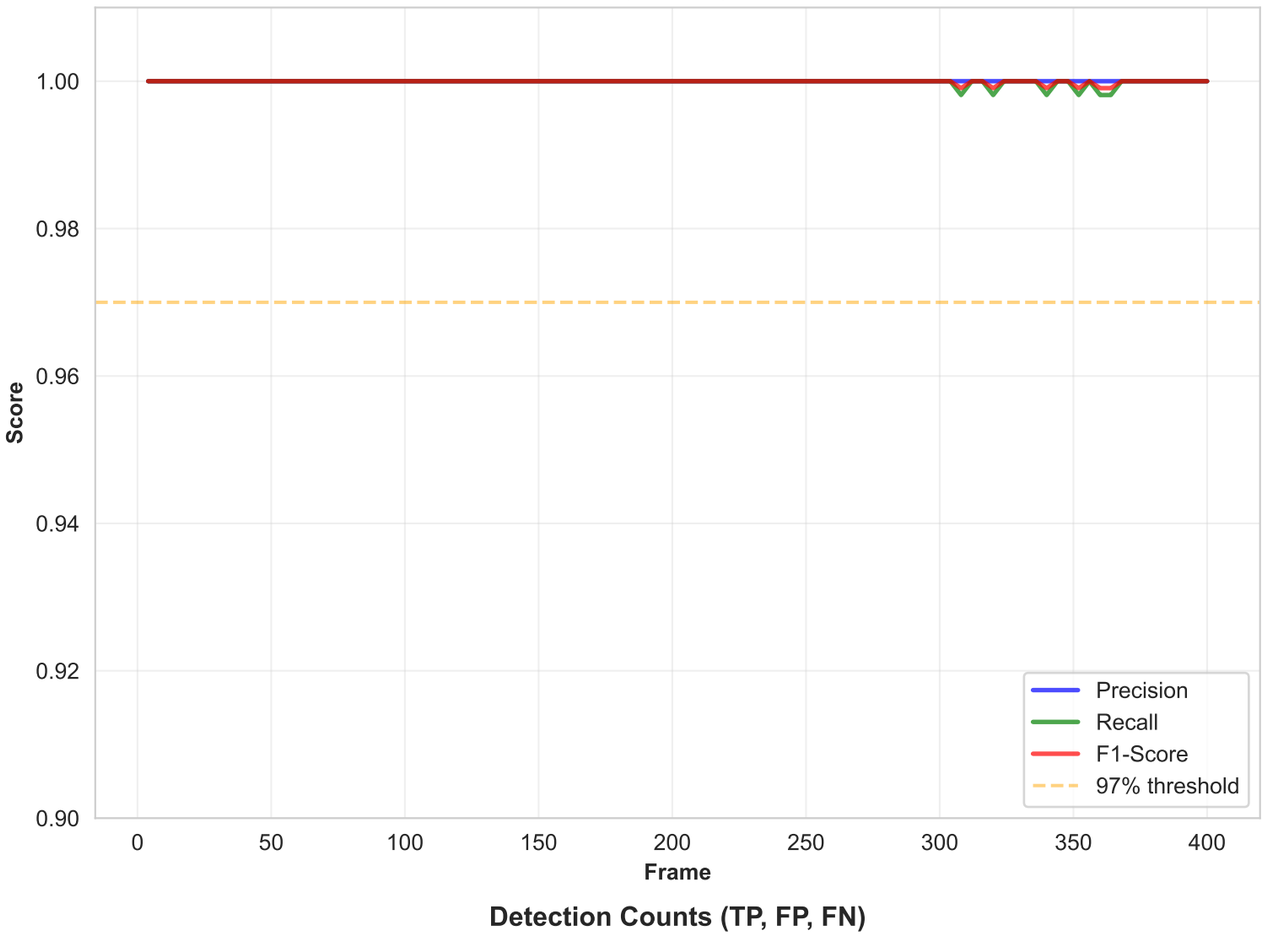
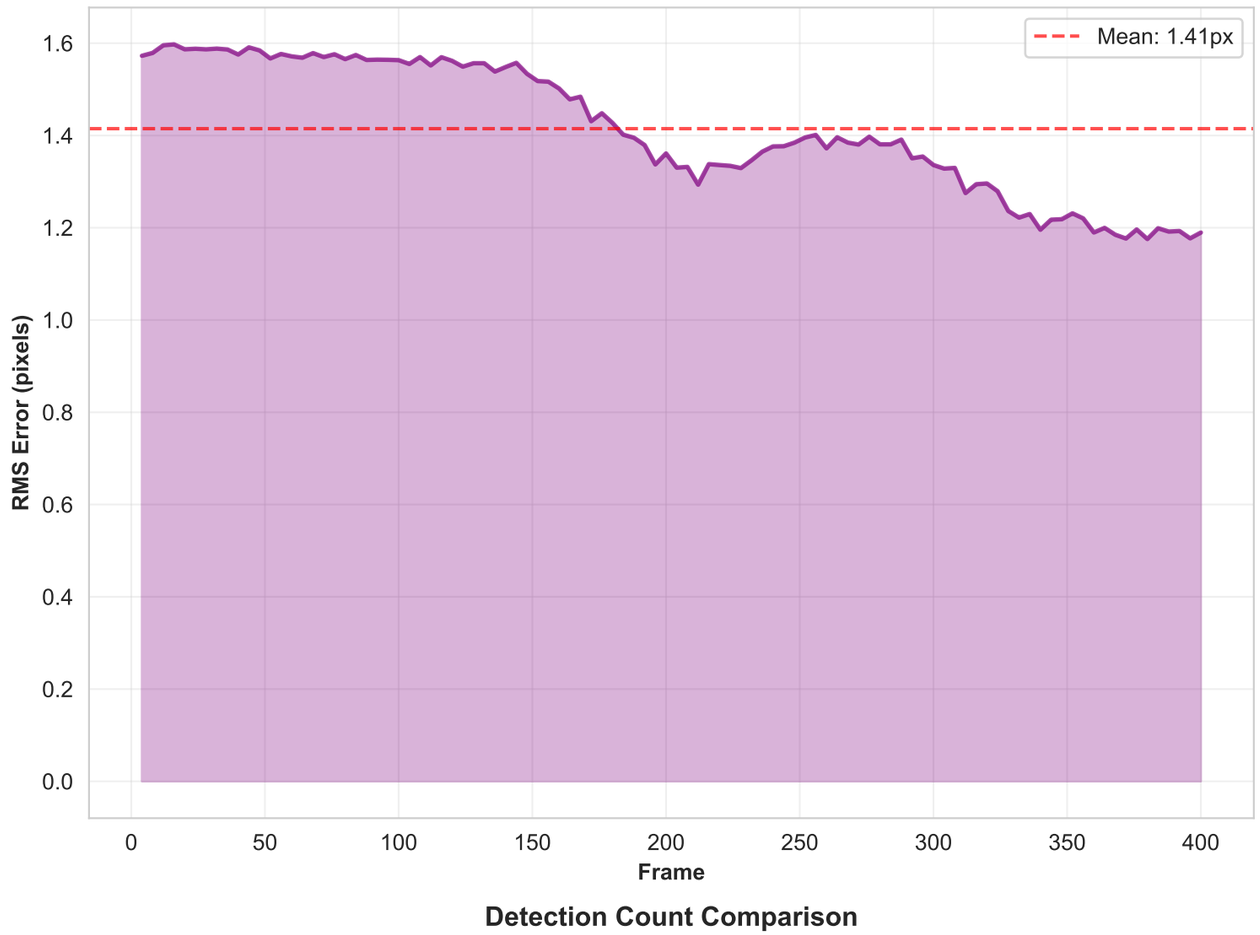
Detection Performance Over Time - tracks_remapped

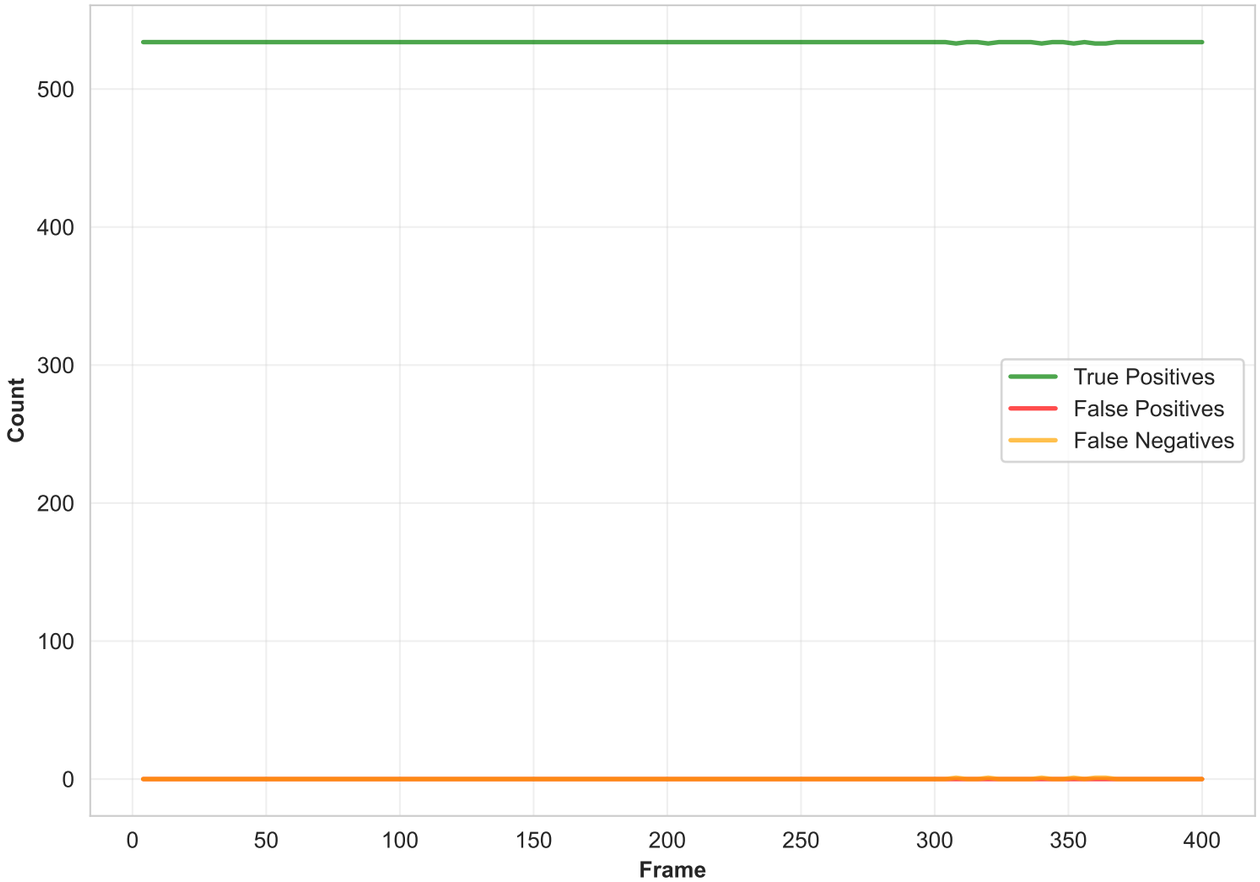
Precision, Recall, F1-Score



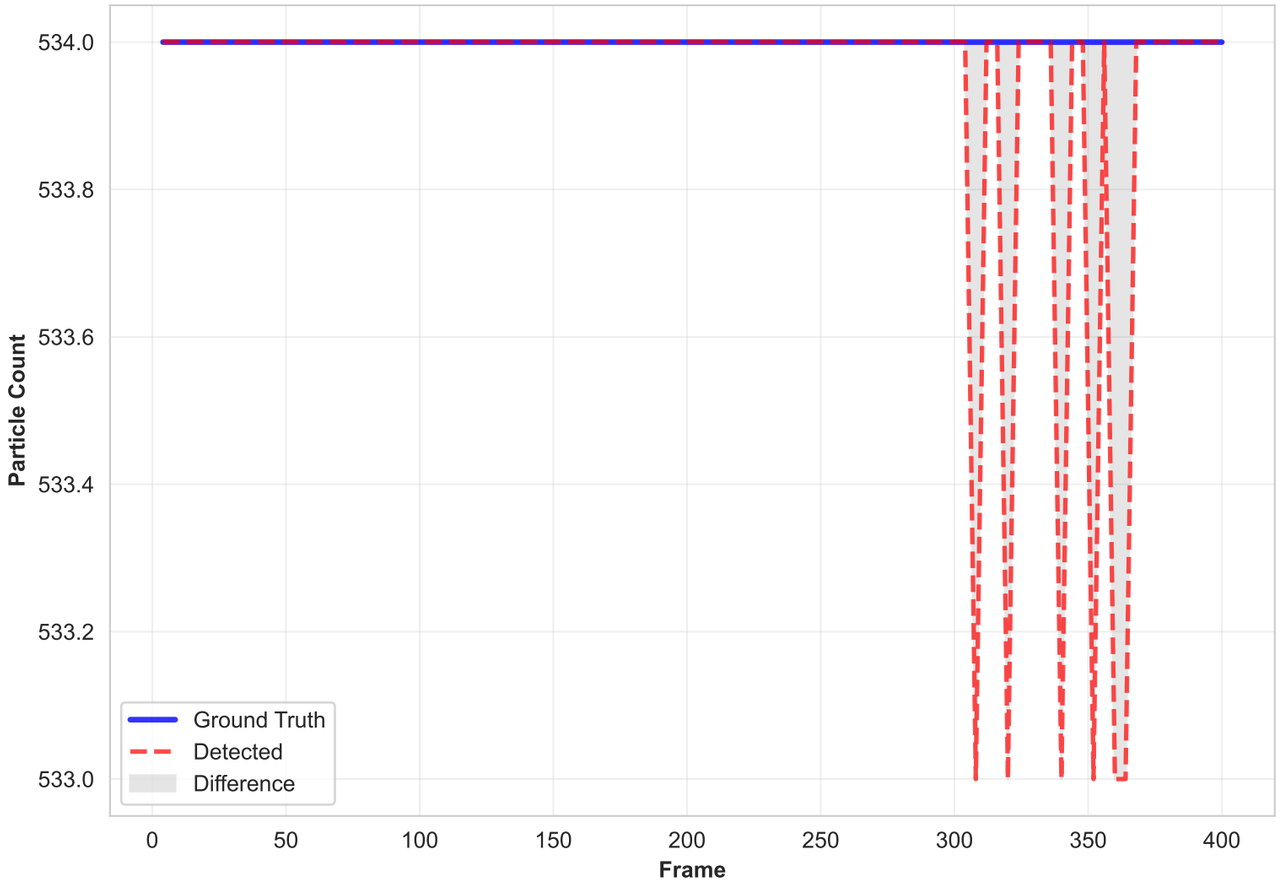
RMS Localization Error



Detection Counts (TP, FP, FN)

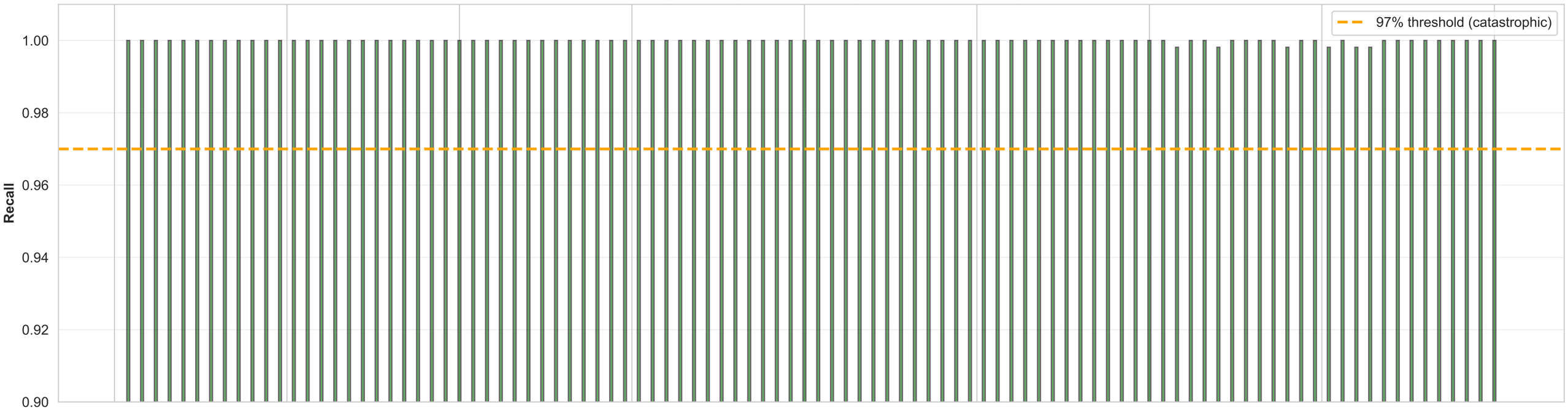


Detection Count Comparison

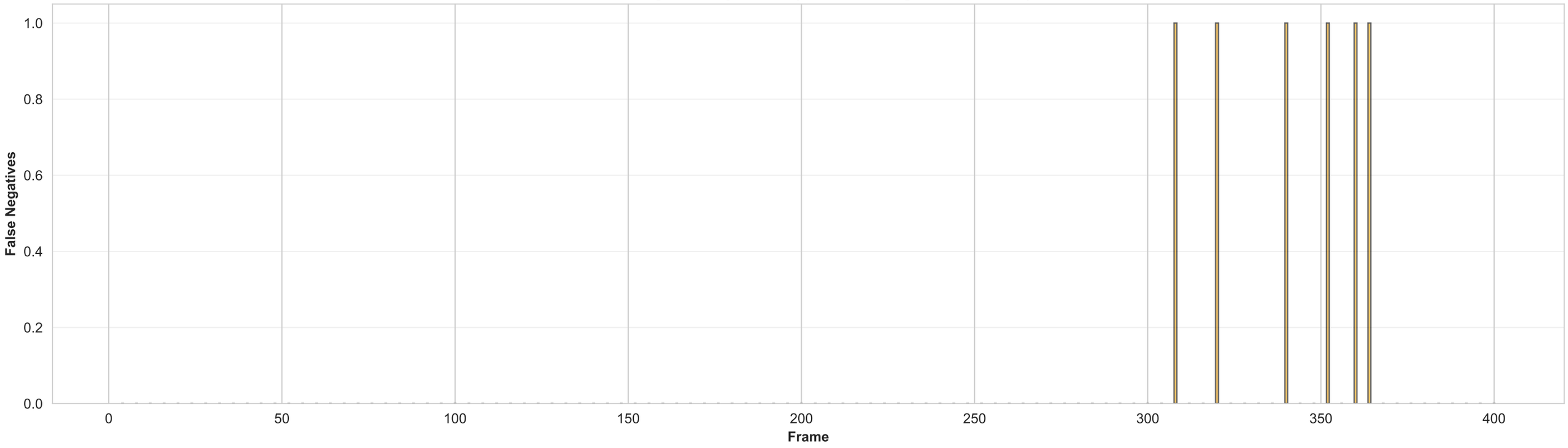


Worst Frames Analysis - tracks_remapped

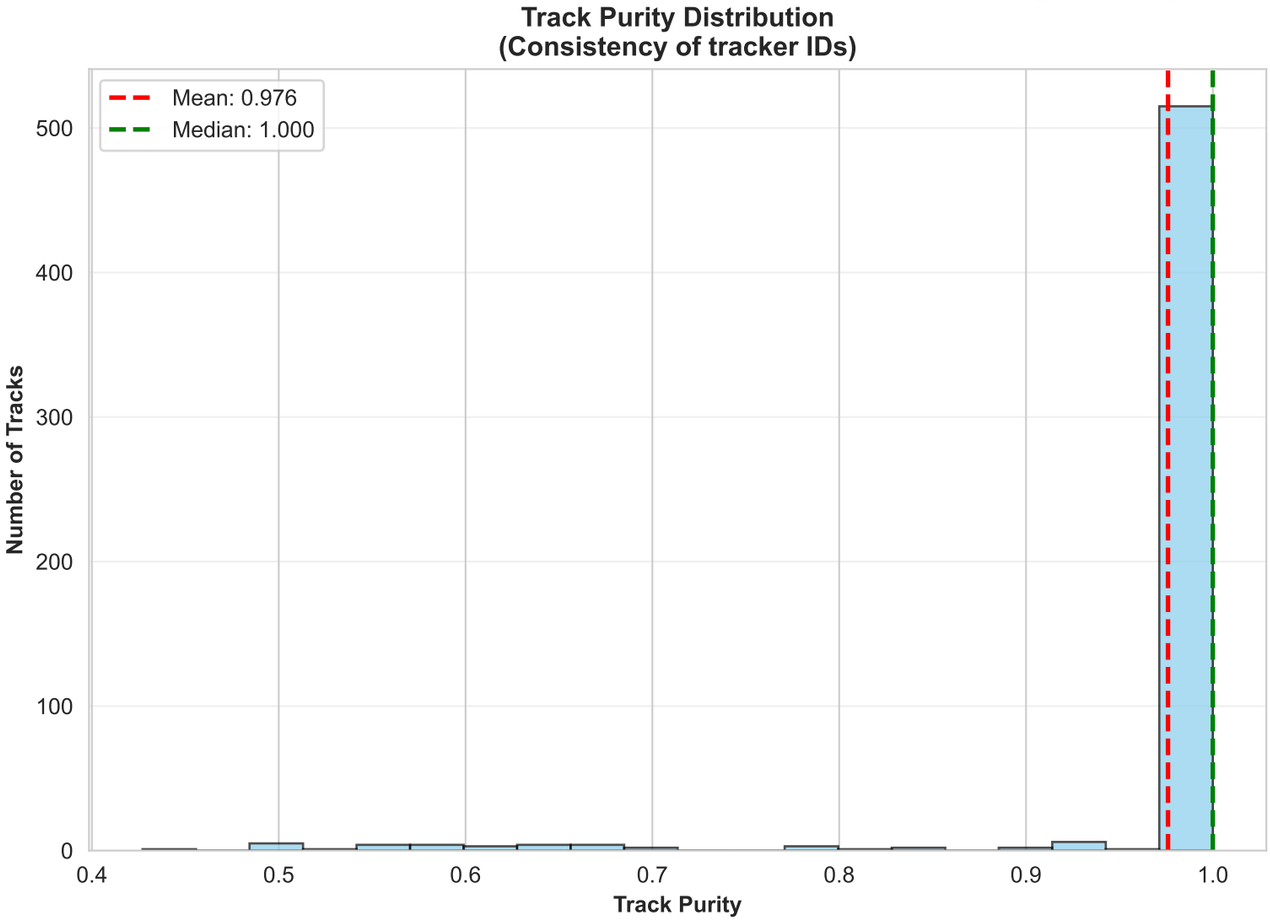
Recall by Frame (Red = catastrophic, 0 frames < 97%)

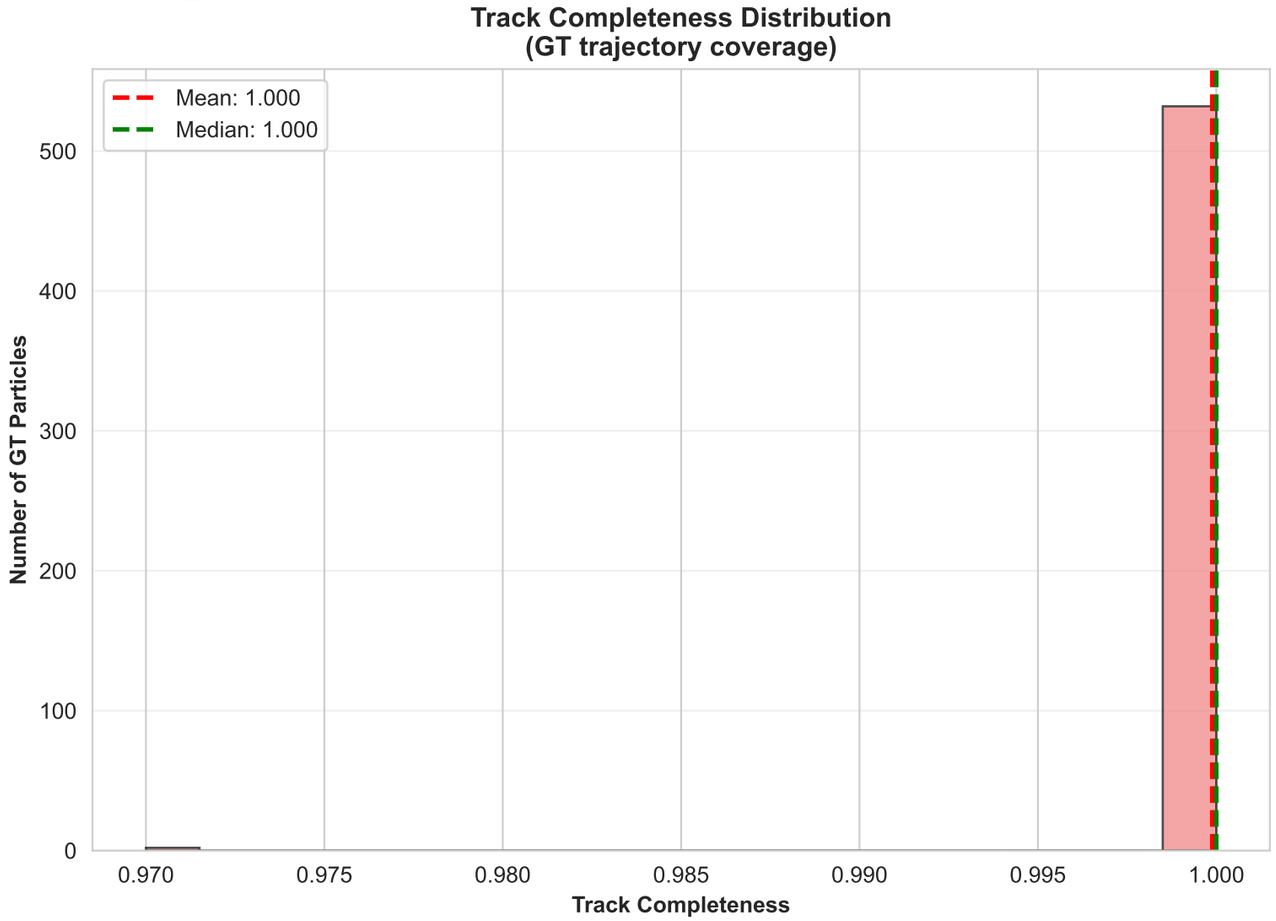


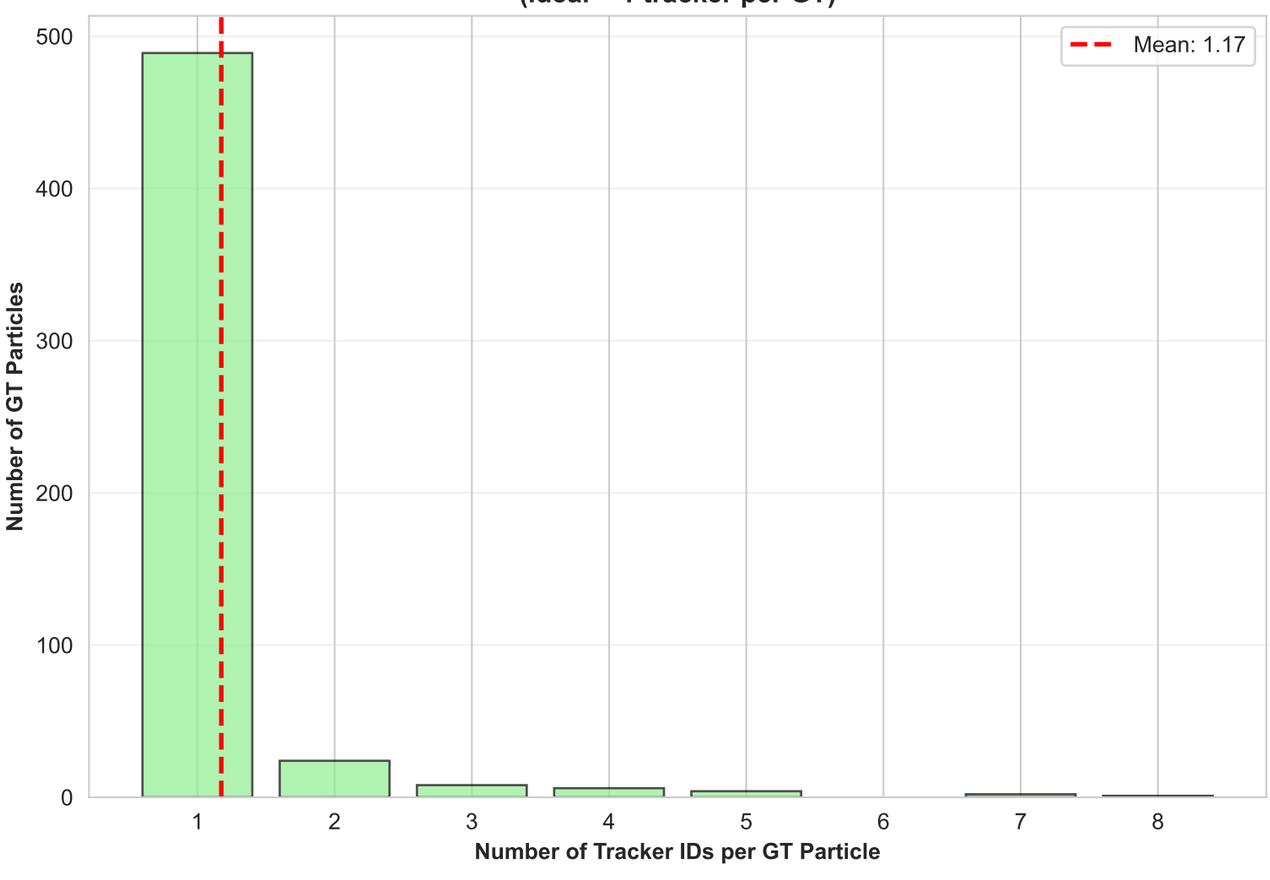
False Negatives by Frame (Red >5, Orange >0, Green =0)

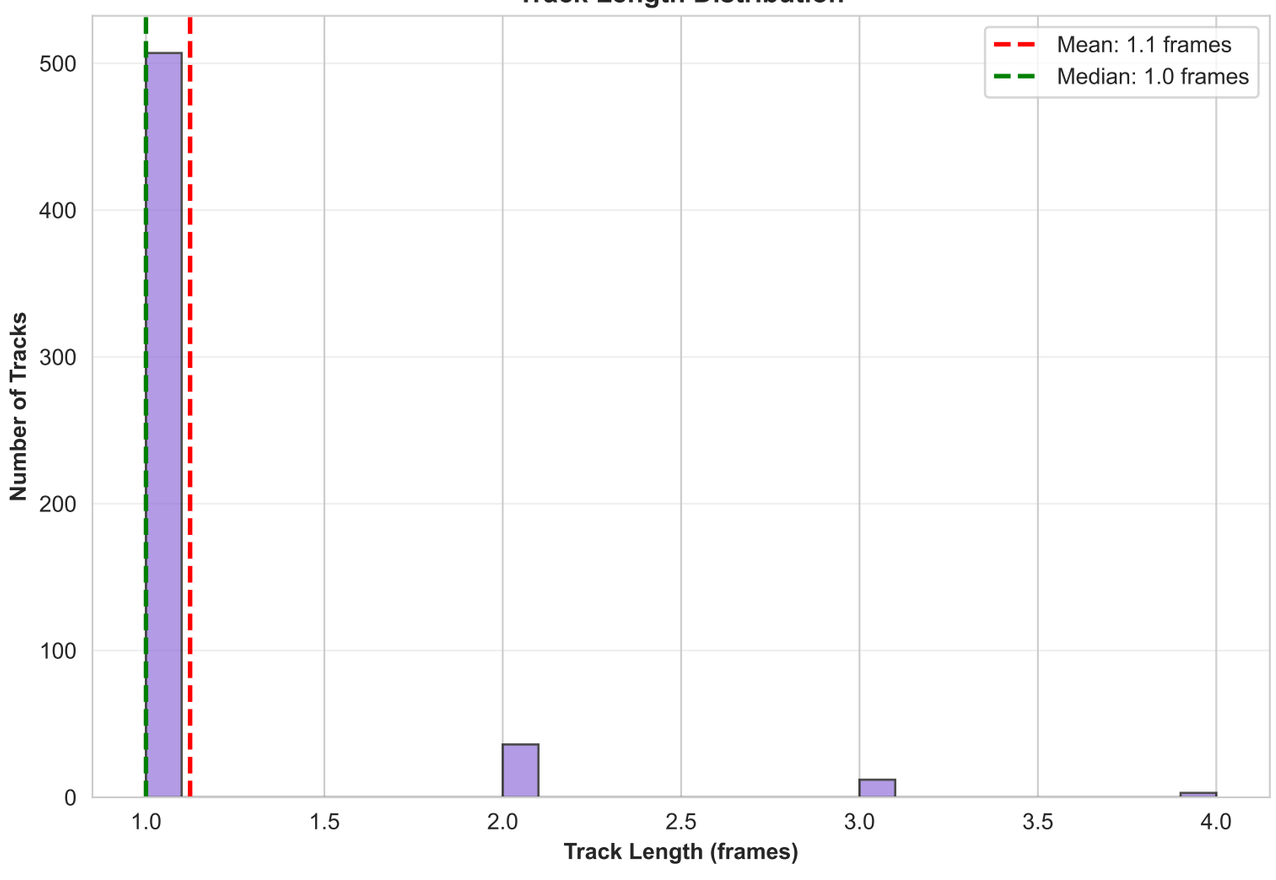


Tracking Quality Metrics - tracks_remapped

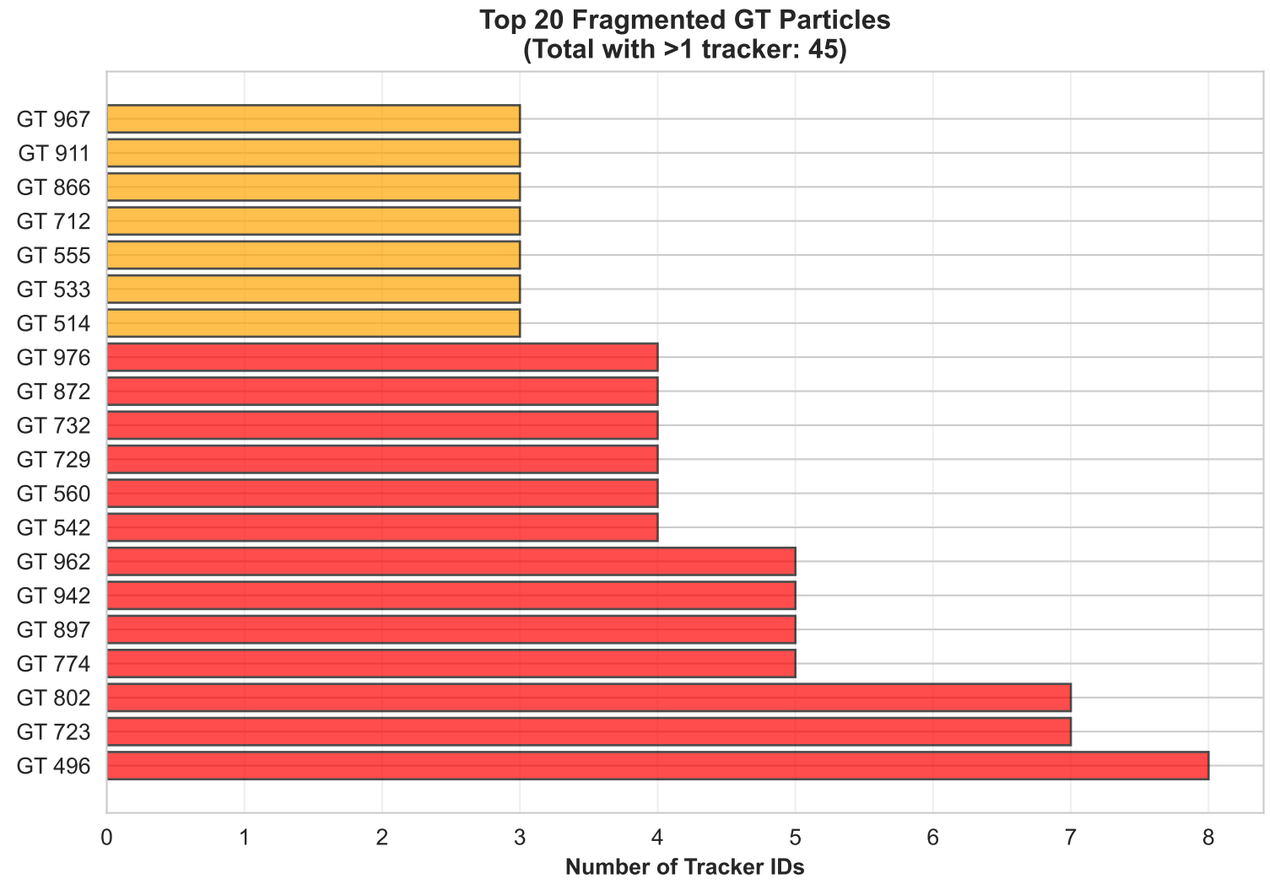


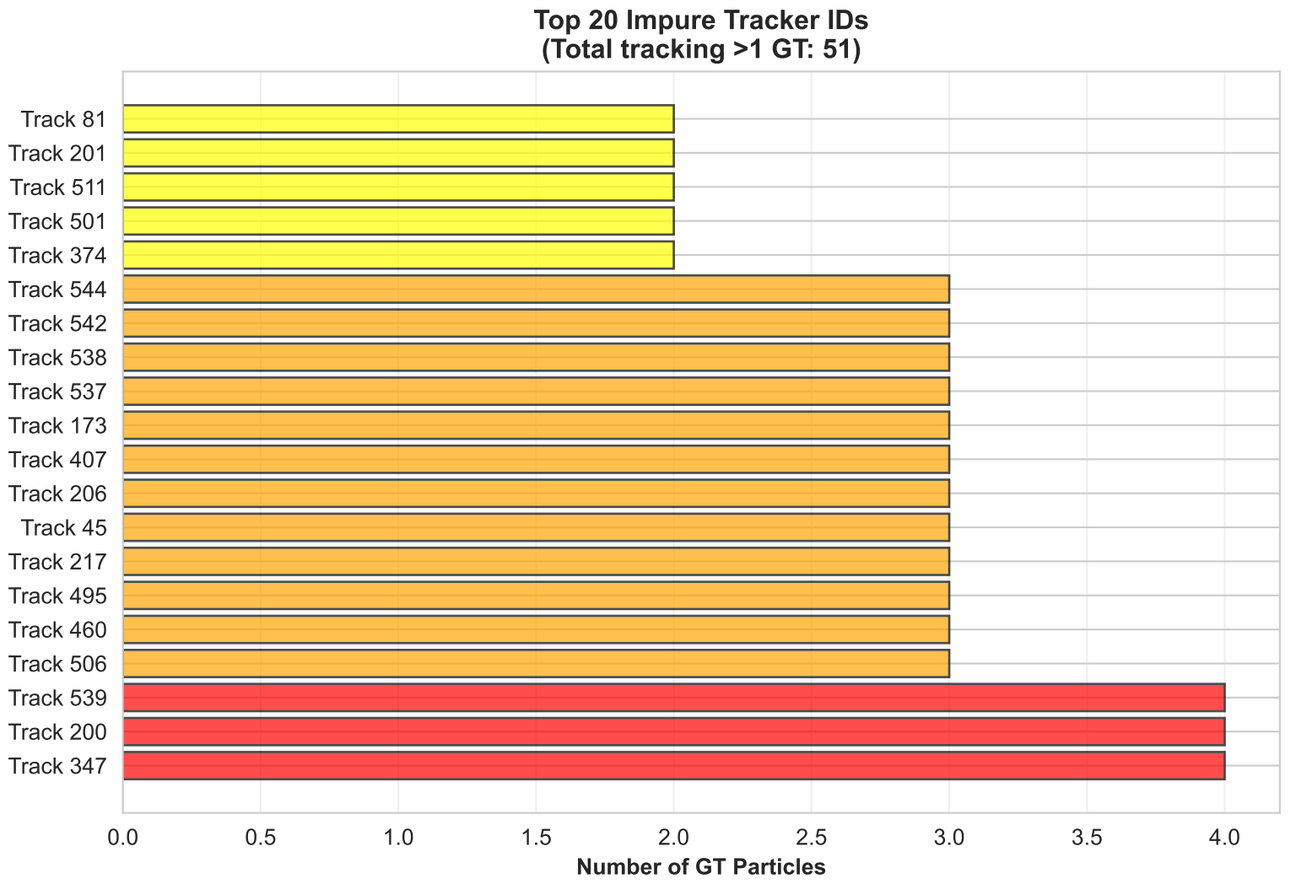




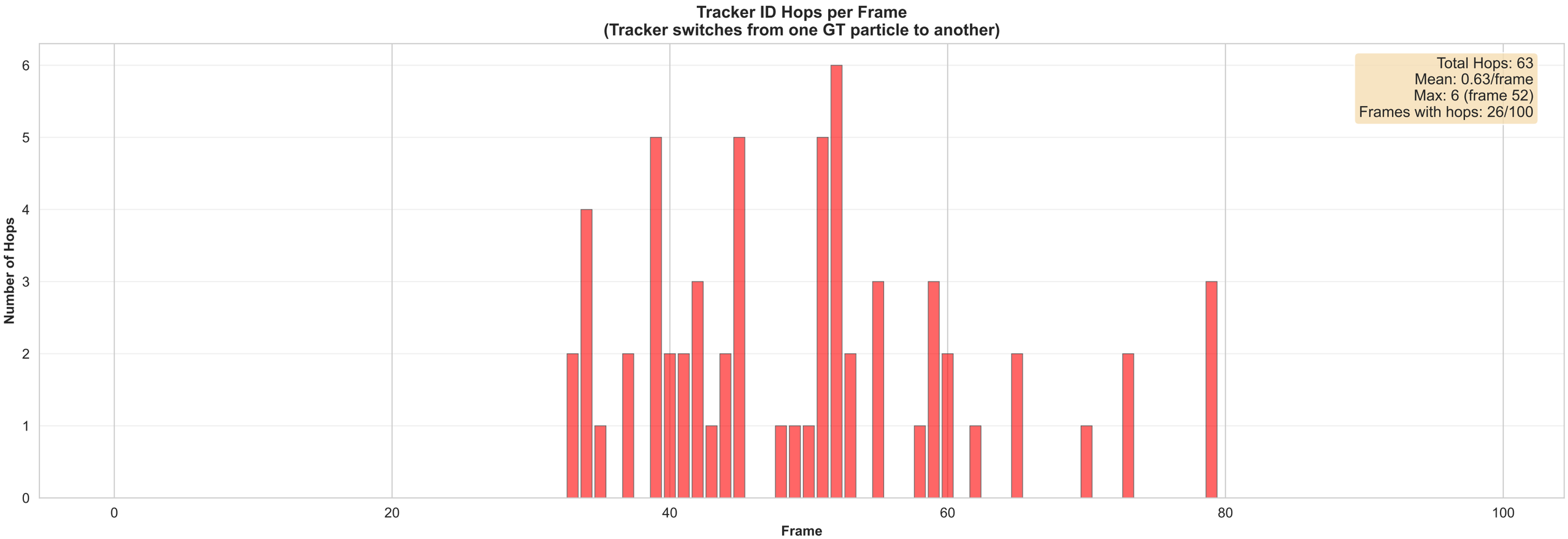


ID Switches and Fragmentation - tracks_remapped





Tracker ID Hops Per Frame - tracks_remapped



Comprehensive Summary - tracks_remapped

```
DETECTION METRICS (Overall)
============================================================
Total Ground Truth Particles:  53,400
Total Detections:              53,394
True Positives:                53,394
False Positives:               0
False Negatives:               6

Precision:                     1.0000
Recall:                        0.9999
F1-Score:                      0.9999

RMS Error:                     1.41 ± 0.14 px

TRACKING METRICS
============================================================
Total GT Particles Tracked:    534
Total Tracker IDs Created:     558
Fragmentation Ratio:           1.04x

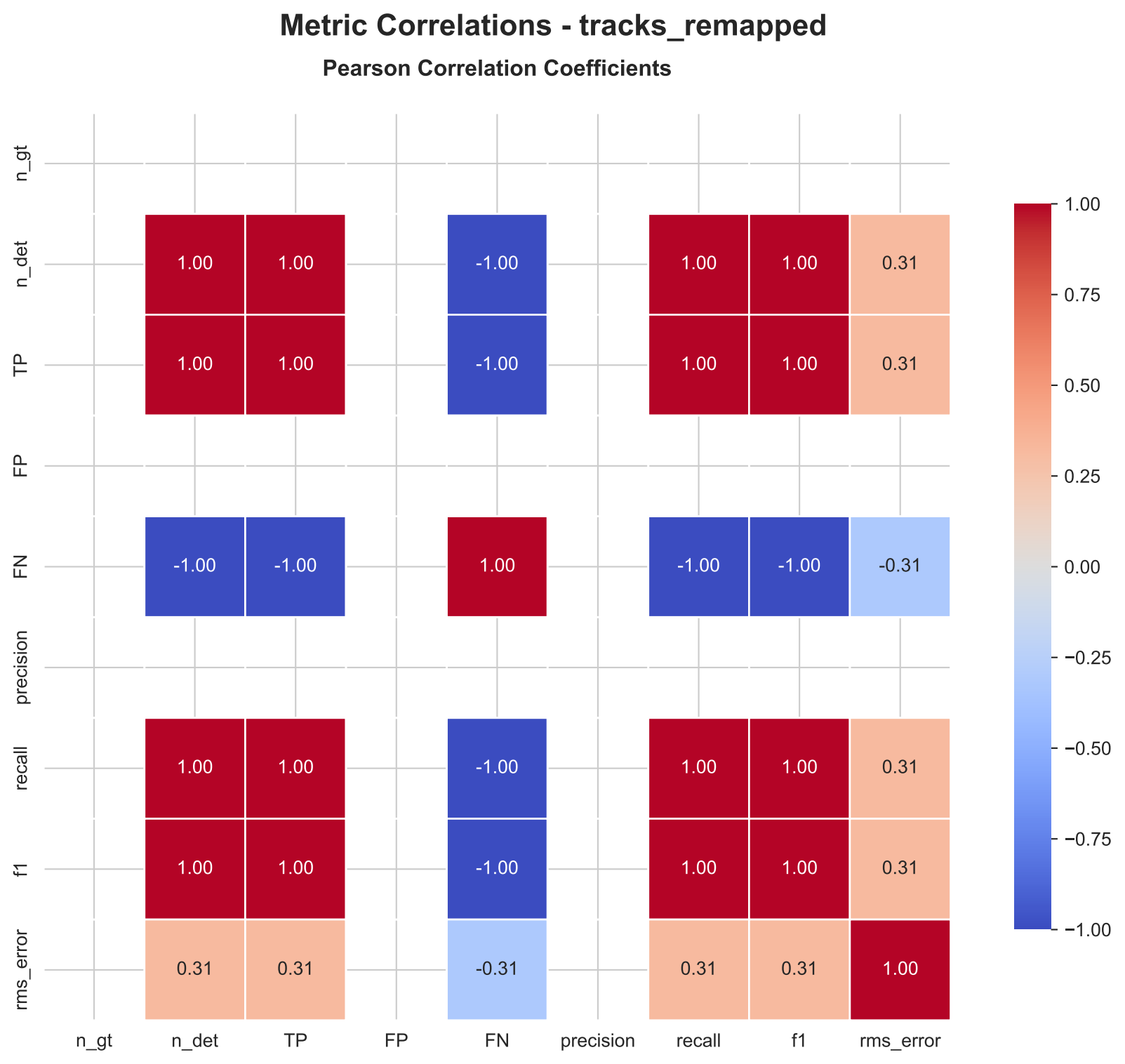
ID Switches (fragmentation):   93
Tracker ID Hops (swaps):       69

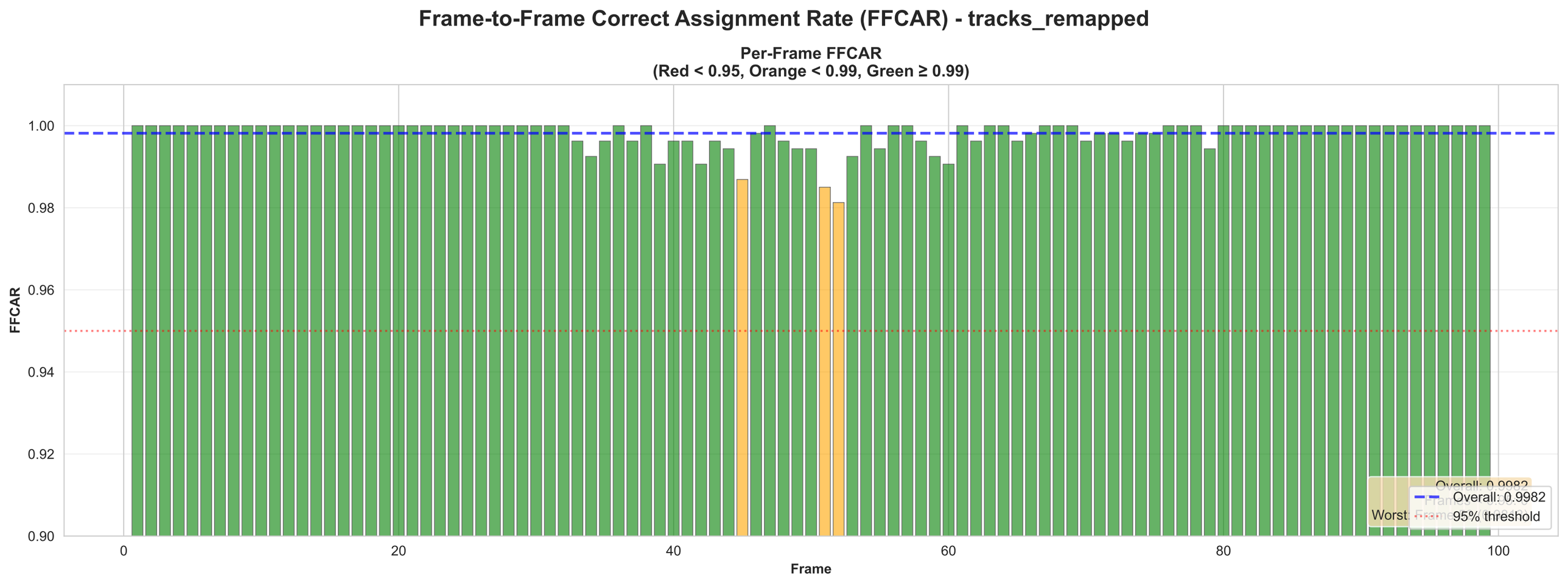
Track Purity (avg/median/min): 0.9760 / 1.0000 / 0.4271
Track Completeness (avg/med):  0.9999 / 1.0000

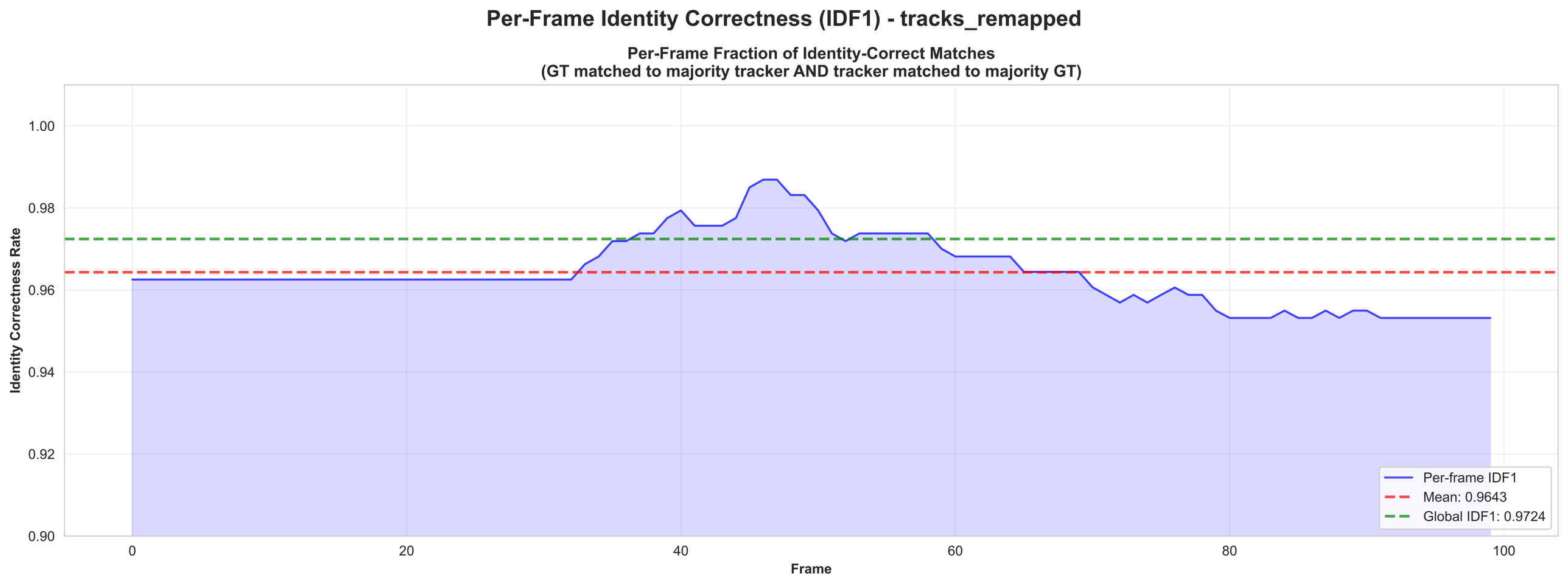
Fragmentation (avg/med/max):   1.17 / 1.0 / 8

MOTA:                          0.9981

WORST FRAMES (Recall < 0.97)
============================================================
None! All frames have recall >= 0.97
```


Metric Correlations - tracks_remapped
Pearson Correlation Coefficients
n_gt
n_det
TP
FP
FN
precision
recall
f1
rms_error
1.00
1.00
-1.00
1.00
1.00
0.31
1.00
1.00
-1.00
1.00
1.00
0.31
-1.00
-1.00
1.00
-1.00
-1.00
-0.31
1.00
1.00
-1.00
1.00
1.00
0.31
1.00
1.00
-1.00
1.00
1.00
0.31
0.31
0.31
-0.31
0.31
0.31
1.00
n_gt
n_det
TP
FP
FN
precision
recall
f1
rms_error
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


Frame-to-Frame Correct Assignment Rate (FFCAR) - tracks_remapped
Per-Frame FFCAR
(Red < 0.95, Orange < 0.99, Green ≥ 0.99)
FFCAR
1.00
0.98
0.96
0.94
0.92
0.90
0
20
40
60
80
100
Frame
Overall: 0.9982
95% threshold


Per-Frame Identity Correctness (IDF1) - tracks_remapped
Per-Frame Fraction of Identity-Correct Matches
(GT matched to majority tracker AND tracker matched to majority GT)
Identity Correctness Rate
1.00
0.98
0.96
0.94
0.92
0.90
0
20
40
60
80
100
Frame
Per-frame IDF1
Mean: 0.9643
Global IDF1: 0.9724